IMDb Top 400 Movies

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Credit: Google Images

Questions

Does a movie’s length affect its rank/rating? Do longer movies tend to score better, or is shorter and tighter the way to go?

Is there a relationship between number of votes (popularity) and rating? Do "popular" movies actually rate higher, or do smaller films sometimes score better?

What genre shows up the most? Has this changed over time?

Credit: Google Images

Variables(Top 400)

Primary Title	**Start Year**	**Rank**	**Average Rating**	**Number of Votes**	**Runtime**	Directors	Writers	**Genres**
The title of the movie	**The year the movie was published**	**IMDb ranking for the movies (1st, 2nd, 3rd, etc.)**	**The average rating given by audience**	**Number of reviews/ votes left by audience**	**The duration of the movie**	The directors of the movie.	The writers of the movie	**The genre of the movie**

Credit: Google Images

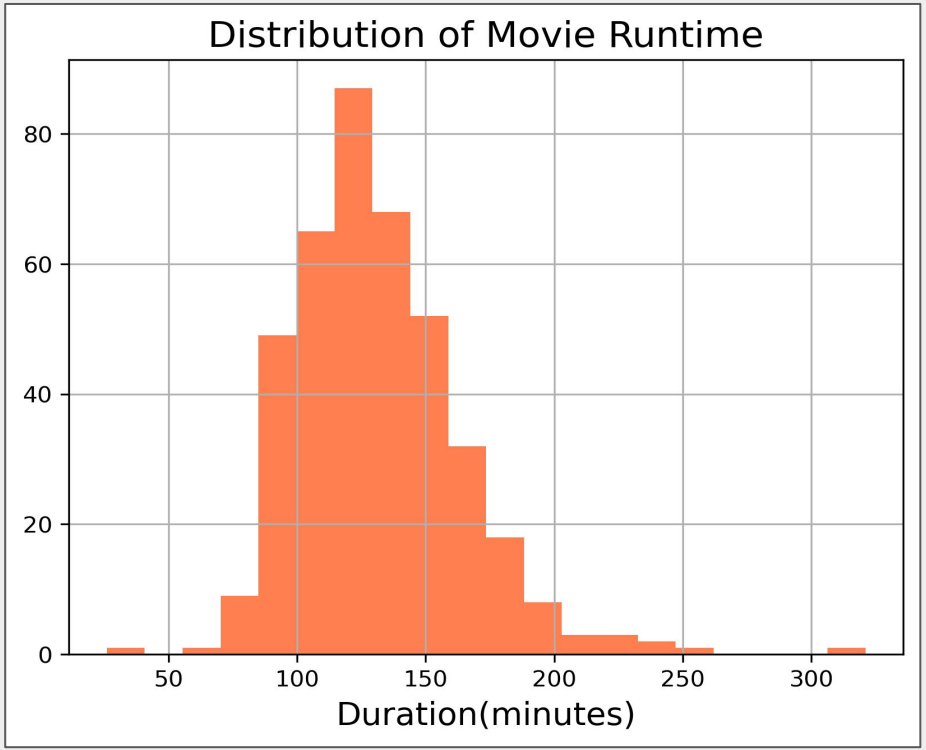

Figure 1: Distribution of movie runtimes. Most films fall between 90 and 150 minutes, with a peak around 110–130 minutes.

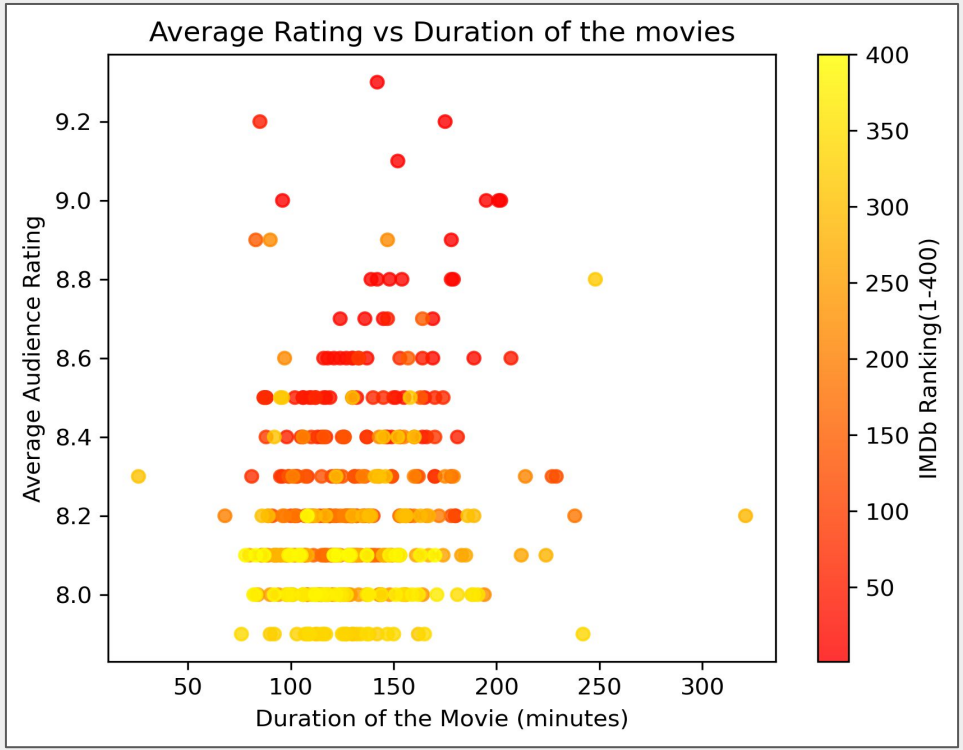

Figure 2: Scatter plot of movie runtime versus average audience rating. The points are widely dispersed, indicating no strong relationship between runtime and rating.

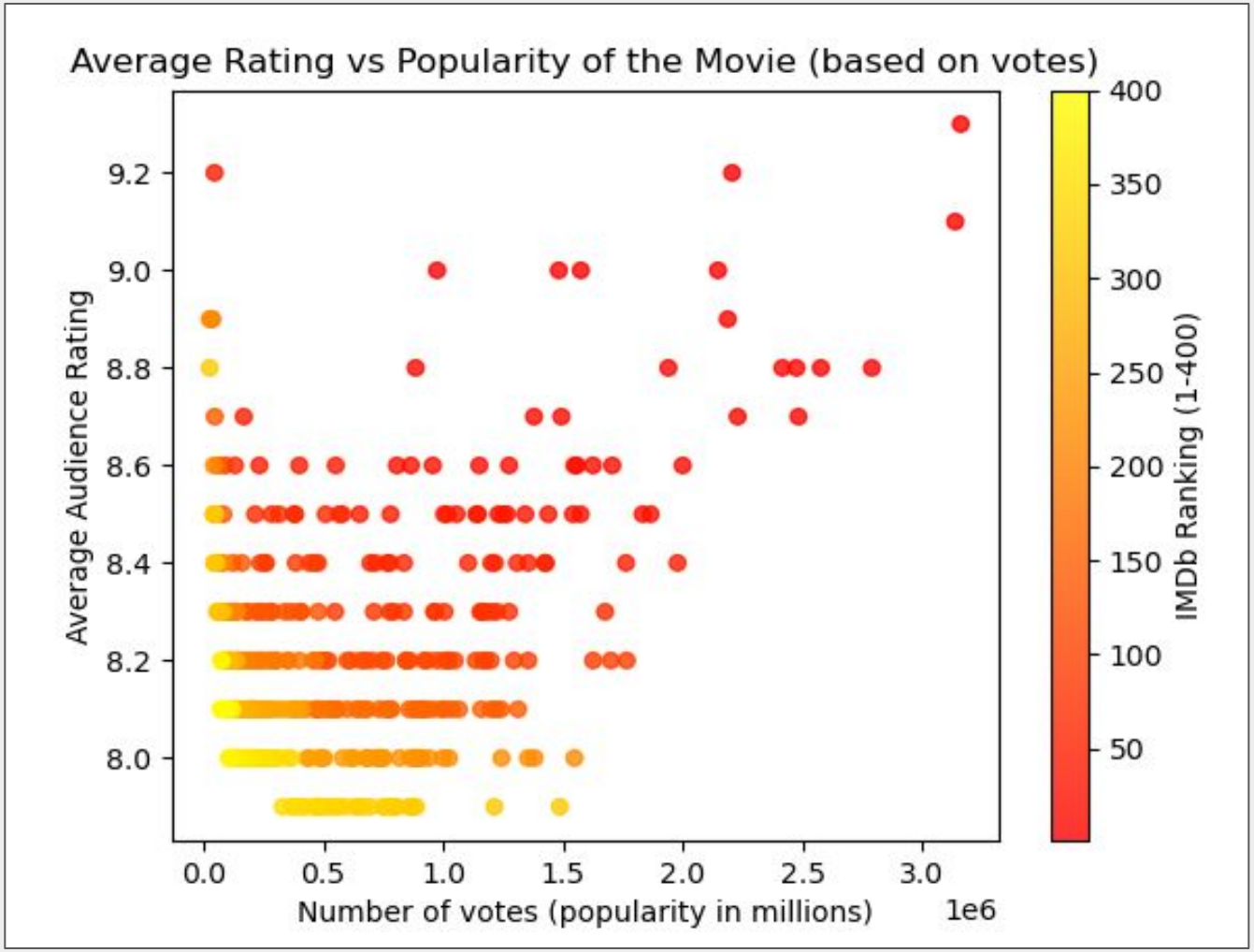

Figure 3: Scatter plot of average audience rating versus number of votes (popularity). The plot shows a slight positive relationship between popularity and rating, as films with a higher number of votes tend to cluster at higher rating values.

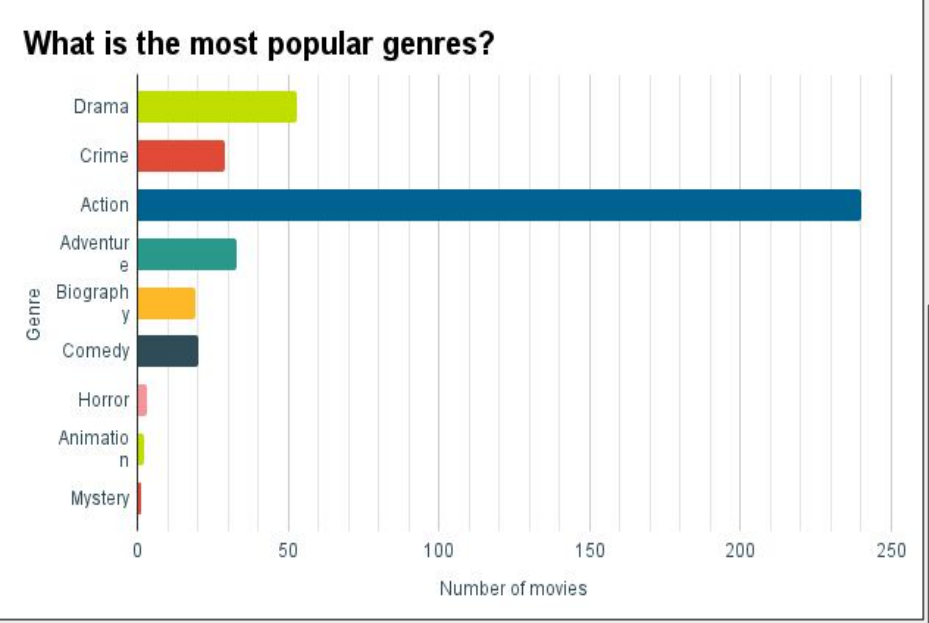

Figure 4: Overall distribution of movie genres in the dataset.

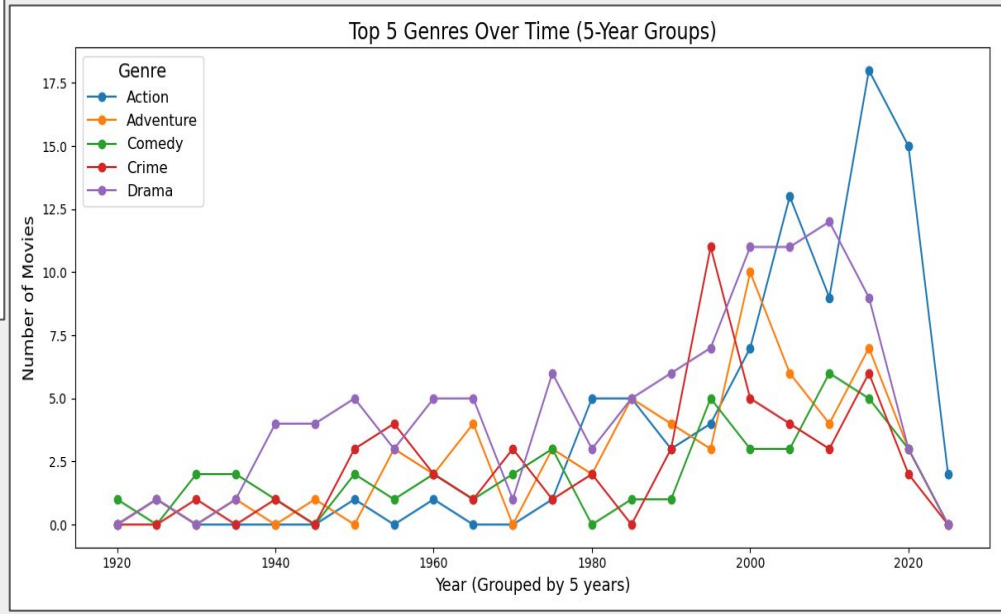

Figure 5: Top five movie genres over time (grouped in 5-year intervals). The graph shows how the number of films in each major genre has changed across decades.

Conclusions & Further Questions

Movie Length & Rating/Rank:	Popularity & Rank/Rate	Genre Quantity Over Time
No strong correlation. Movie length does not affect a movie's rank.	Slight positive correlation. More popular movies tend to rank higher.	The most popular genre changes over time. Action is currently most popular.

Questions: What variables of a movie affect its ranking the most? What matters most to people's enjoyment of a movie? Has this changed over time?

Credit: Google Images

Thank you!

Credit: Google Images

Sources

Nunes, Tiago Adria. “IMDb Top 5000 Movies.” *Kaggle.com*, 2025, www.kaggle.com/datasets/tiagoadrianunes/imdb-top-5000-movies.